

Projection Normalization: Before vs After

Before Normalization (INCORRECT)

[protocol]

[dst_ip] 10.0-10.8

[dst_ip] 10.3-10.10

overlap at 10.3-10.8

Packet dst = 10.0.0.4 → matches BOTH nodes

✗ Ambiguous: sibling ranges overlap
— disjointness invariant not met

↓ Algorithm 1

After Normalization (CORRECT)

[protocol]

[dst_ip]
10.0-10.2

[dst_ip]
10.3-10.8

[dst_ip]
10.9-10.10

disjoint sub-ranges (no overlap)

Packet dst = 10.0.0.4 → matches EXACTLY ONE rule

✓ Unambiguous: sibling ranges disjoint
— invariant holds (Corollary 1, Theorem 2)